

Omar Ahmad Elhakim

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Education

Ain Shams University

BSc in Computer Science, Scientific Computing Department | GPA: 3.3

Graduated June 2026

Experience

Zinad

Software Engineer (Part-Time)

July 2024 – Present

- Built a phishing simulation engine that generates themed emails and working fake landing pages from the parameters a user picks, driving **Ollama** from a **Laravel** backend and rendering the emails with **MJML**.
- Wrote an email analyzer in **Python**, **LangChain** and **Ollama** that flags phishing from tone, context and grammatical anomalies; its verdicts feed the platform's security metrics.
- Developed an internal **RAG** system that answers employee questions from the company's policy PDFs, splitting the documents with **recursive character chunking** with overlap, embedding the chunks into a **FAISS** vector store and retrieving the passages relevant to each question.
- Shipped a chatbot into the live application with **LangChain** and **Ollama**, streaming replies token by token instead of making the user wait for the full answer.

Projects

RAG

- Built a document Q&A service in **FastAPI** and **LangServe**, splitting a company policy corpus into overlapping chunks with a **recursive character splitter**, embedding them with **word2vec**, indexing them in a **Chroma** vector store, and grounding **Gemini** on the chunks retrieved for each question.
- Added a phishing screener that runs an email body through a self-hosted **Llama** on **Ollama**, returning nine checks (urgency language, requests for confidential information, character-encoding tricks such as Cyrillic lookalikes, and others) as structured **JSON** with English and Arabic justifications, covered by end-to-end API tests.
- Extended retrieval to databases with a **text-to-SQL** chain that feeds the live **MySQL** schema to the model, executes the generated query and phrases the result in plain English.

Encryption and Cryptanalysis Toolkit

- Implemented ten ciphers from scratch in **Java** with no cryptography libraries, from classical ones (Caesar, Playfair, Hill, Vigenère, rail fence, columnar transposition) up to full **AES-128** and **DES** block ciphers, including the byte substitution, row shifts, column mixing and key expansion.
- Broke the classical ciphers as well as building them, recovering the key from a known plaintext and ciphertext pair, and cracking the monoalphabetic cipher by letter-frequency analysis alone, all covered by **45 JUnit tests**.

Graph-Based Image Segmentation

- Split images into regions in **C#**, implementing a **graph segmentation** algorithm from its paper over a graph built from every pixel.
- Sorted those edges in **linear time** with a 256-bucket **counting sort** rather than an $O(E \log E)$ comparison sort.
- Segmented the red, green and blue channels independently and intersected the three results.

Pac-Man with Pathfinding Ghosts

- Built a Pac-Man clone in **C++** and **raylib** with four ghosts that each run a **breadth-first search** over the maze grid and step along the shortest path; difficulty changes what they search for, from a random corridor to Pac-Man himself.
- Loaded the mazes from **CSV** files and shipped a mouse-driven map editor that lets a player draw a maze and immediately play it, with persisted high scores and a resizable window that rescales the grid.

Technical Skills

Languages: Python, C/C++, C#, Java, PHP, JavaScript, SQL, Bash

Tools/Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, OpenCV, LangChain, FastAPI, Laravel, Docker, Git, Linux, Django

Training & Development

- **MEC Model of Leadership:** Completed an extensive leadership program (160 hours) covering emotional intelligence, effective communication, self-awareness (50 hours), the Enneagram, problem-solving, and team building.